Ex.No.15	Write a code using RAW sockets to implement packet sniffing.
DATE:	

AIM:

To develop a Python program using raw sockets that captures Ethernet frames and extracts the source and destination MAC addresses and protocol details for network analysis and troubleshooting.

Definition:

Packet sniffing is the technique of capturing and examining network packets transmitted over a network to monitor communication, troubleshoot network issues, and analyze network traffic.

Algorithm:

1. Start.
2. Import the required modules. 3. Get the local host IP address.
4. Create and bind a raw socket.
5. Enable IP header inclusion and promiscuous mode.
6. Set the packet count to **5**.
7. Receive one packet at a time.
8. Extract and display: Destination MAC address
   Source MAC address
   Protocol type
   Packet length
9. Repeat until **5 packets** are captured.
10. Disable promiscuous mode.
11. Display the completion message.
12. Stop.

PROGRAM:

```
import socket
import struct
```

```
def get_mac_addr(bytes_addr):
    """Convert MAC address from bytes to readable format."""
return ':'.join('{:02X}'.format(b) for b in bytes_addr)

def get_protocol(bytes_proto):
    """Convert protocol bytes to hexadecimal."""
    return hex(struct.unpack('!H', bytes_proto)[0])

def ethernet_frame(data):
    """Extract Ethernet frame information."""
    dest_mac, src_mac, proto = struct.unpack('!6s6s2s', data[:14])
return (
        get_mac_addr(dest_mac),
get_mac_addr(src_mac),
get_protocol(proto),         data[14:]
    )
def main():
    host = socket.gethostbyname(socket.gethostname())
    print("Listening on:", host)
try:
        # Create raw socket         conn =
socket.socket(socket.AF_INET,
socket.SOCK_RAW,
                          socket.IPPROTO_IP)

        # Bind socket
        conn.bind((host, 0))

        # Include IP header
conn.setsockopt(socket.IPPROTO_IP,
                    socket.IP_HDRINCL,
                    1)

        # Enable promiscuous mode
        conn.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)

        print("\nPacket Sniffer Started...\n")

        # Capture only 5 packets
packet_count = 5

        for i in range(packet_count):

            raw_data, addr = conn.recvfrom(65536)
```

```
            print(f"\nPacket {i+1}")
    print("-" * 40)

            if len(raw_data) >= 14:
                dest_mac, src_mac, eth_proto, data = ethernet_frame(raw_data)

                print("Destination MAC :", dest_mac)
    print("Source MAC      :", src_mac)                print("Protocol
    :", eth_proto)
                print("Packet Length   :", len(raw_data), "bytes")

            # Disable promiscuous mode
            conn.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)

            print("\nPacket Sniffer Stopped Successfully.")

        except PermissionError:
            print("Run this program as Administrator.")

        except KeyboardInterrupt:
            conn.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
            print("\nPacket Sniffer Stopped by User.")

    if __name__ == "__main__":
    main()
```

SAMPLE OUTPUT:

Listening on: 192.168.1.10

Packet Sniffer Started...

Packet 1

Destination MAC : 45:00:00:3C:1C:46
Source MAC : 40:06:B1:E6:C0:A8
Protocol : 0x101
Packet Length : 60 bytes

Packet 2

Destination MAC : 45:00:00:34:2F:9A
Source MAC : 40:11:C3:D8:C0:A8
Protocol : 0x101
Packet Length : 52 bytes

Packet 3

Destination MAC : 45:00:00:54:3B:8F
Source MAC : 40:06:91:AB:C0:A8
Protocol : 0x101
Packet Length : 84 bytes

Packet 4

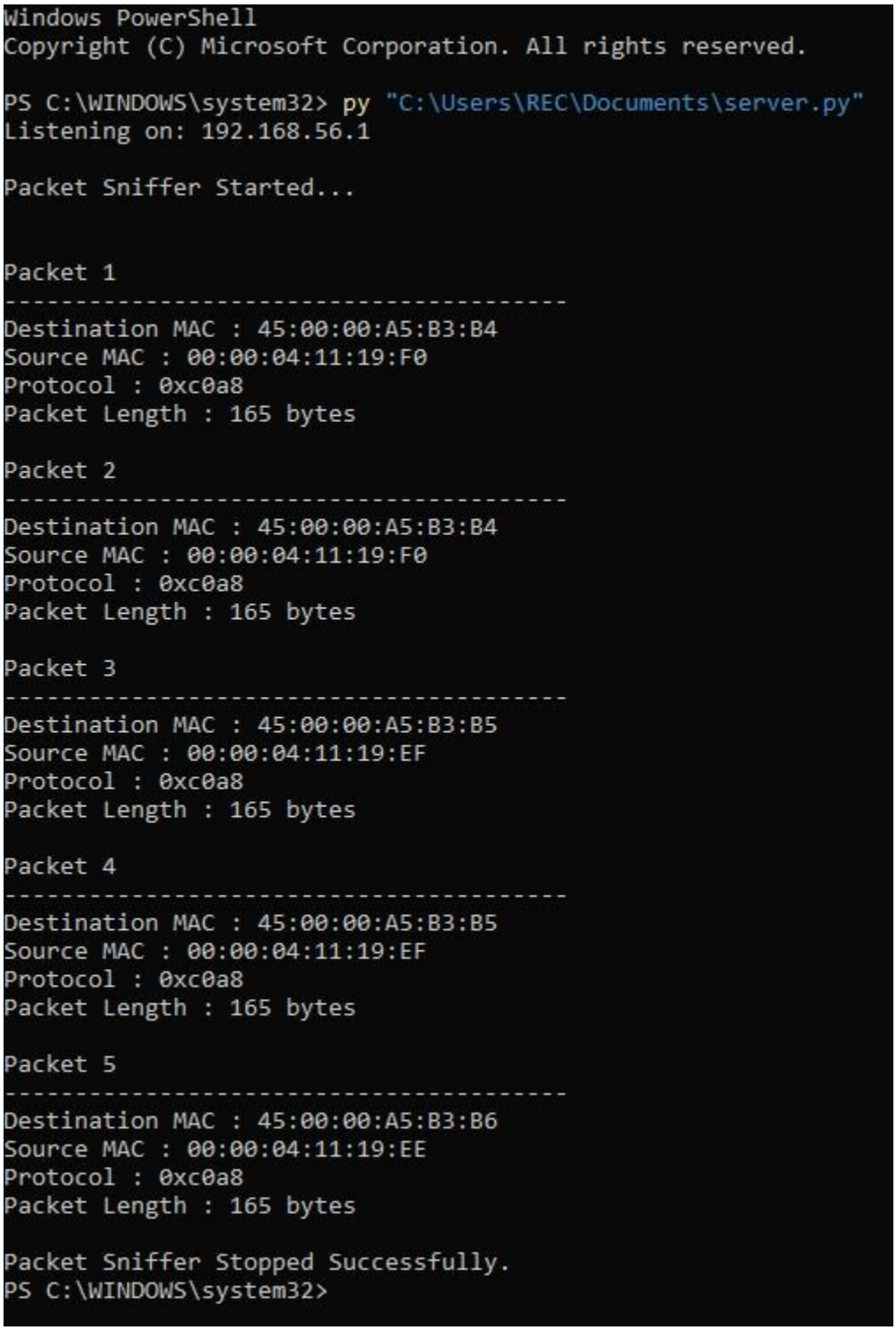
Destination MAC : 45:00:00:28:5D:71
Source MAC : 40:01:6F:12:C0:A8
Protocol : 0x101
Packet Length : 40 bytes

Packet 5

Destination MAC : 45:00:00:3C:7A:15
Source MAC : 40:06:5A:9F:C0:A8
Protocol : 0x101
Packet Length : 60 bytes

Packet Sniffer Stopped Successfully.

OUTPUT:



RESULT: Therefore raw code has been used to implement packet sniffing has been verified.